

Coherent transformation of metal halide perovskites

Received: 4 January 2026

Accepted: 24 June 2026

Published online: 10 July 2026



Pinghui Yang^{1,8}, Haokun Shi^{1,8}, Zhen Wang^{1,8}, Shuaijun Yan¹, Xiliu Wang¹, Bo Xu¹, Junbo Wang¹, Kaidi Liu¹, Xuan Gao¹, Wenbo Liu^{1,2}, Fangfang Wang¹, Lin Zhu¹, Renzhi Li¹✉, Wei Huang^{1,3,4,5}✉ & Jianpu Wang^{1,6,7}✉

Metal-halide perovskites offer high optoelectronic performance but translating laboratory efficiencies into controllable, reproducible processes remains challenging. Perovskite crystallization is tracked by in situ multiwavelength absorption to show that coherent, layer-by-layer conversion of low-*n* intermediates to three-dimensional perovskite, quantified by a transformation index (TI), correlates with device quality. TI displays a Sabatier-type dependence on additive–precursor interaction strength and concentration, defining an engineering window for reproducible film formation. Within this window, post-treatment-free devices reach a champion power conversion efficiency of 26.40% with improved stability and reduced device scatter at laboratory scale. TI is extractable from an inline optical readout and provides a measurable metric for additive screening and process tuning. Its applicability to scalable deposition is further validated through slot-die coated mini-modules, offering a practical pathway for inline process control.

Metal-halide perovskite semiconductors, with their remarkable optoelectronic properties, have emerged as promising materials for a wide range of applications, including solar cells, light-emitting diodes, and photodetectors^{1–5}. Turning laboratory efficiencies into controllable, reproducible processes remains a central challenge, since approaches that deliver best laboratory performance frequently show poor reproducibility between batches and under different deposition conditions⁶. Solution processing of metal-halide perovskites requires precise control of rapid crystallization kinetics, including solvent-coordinated complexes^{7,8}, transient low-dimensional intermediates⁹, nucleation density, and subsequent grain growth¹⁰, so that solvated precursors reproducibly convert into phase-pure, low-defect three-dimensional lattices across different coating platforms and scales. A broad set of approaches has been developed to address these issues,

including additive engineering^{11–20}, solvent and antisolvent protocols^{21–24}, single-crystal redissolution^{25,26}, and low-temperature crystallization^{27–29}. However, these methods are often treated as separate strategies rather than as a common nucleation-control principle. As a result, there is no practical, quantitative process-control paradigm that connects precursor chemistry and crystallization dynamics to scalable processing windows, inline monitoring metrics, and reproducible device yields across coating platforms.

Here we show that a coherent transformation, defined as a temporally ordered, sequential layer-by-layer evolution from transient low-dimensional (layered) intermediates into the final 3D perovskite, is the common microstructural signature of reproducible, low-defect films. This mechanism is not unique to additives: it also emerges when using single-crystal redissolution (SCR) and low-temperature cry

¹Key Laboratory of Flexible Electronics (KLOFE), Institute of Advanced Materials (IAM) & School of Flexible Electronics (Future Technologies), Nanjing Tech University (Nanjing Tech), Nanjing 210003 Jiangsu, China. ²Strait Institute of Flexible Electronics (SIFE, Future Technologies), Fujian Normal University, Fuzhou 350117 Fujian, China. ³Institute of Flexible Electronics (IFE), Northwestern Polytechnical University (NPU), Xi'an 710072 Shaanxi, China. ⁴Strait Laboratory of Flexible Electronics (SLoFE), Fuzhou 350117 Fujian, China. ⁵School of Flexible Electronics (SoFE), Sun Yat-sen University, Shenzhen 518107 Guangdong, China. ⁶National & Local Joint Engineering Research Center of Technical Fiber Composites for Safety and Health, School of Textile and Clothing, Nantong University, Nantong 226019, China. ⁷The Materials and Electronics Research Center (MERC), School of Materials Science and Engineering, School of Microelectronics and Control Engineering, Changzhou University, 21 Middle Gehu Road, Changzhou 213164, China. ⁸These authors contributed equally: Pinghui Yang, Haokun Shi, Zhen Wang. ✉e-mail: iamrzli@njtech.edu.cn; vc@nwpu.edu.cn; iamjpwang@njtech.edu.cn

stallization, which each establish a single, ordered nucleation pathway that guides sequential phase evolution rather than overlapping, disordered transitions. To quantify this behavior, we introduce a transformation index (TI) derived from high-resolution, time-resolved absorption spectra; TI captures the temporal ordering of intermediate formation and conversion and serves as an inline, process-readable metric. Using TI as a control parameter, we find a Sabatier-type (volcano) dependence of device performance and defect density on additive-precursor interaction strength and concentration, defining an engineering window with practical tolerance. Operating inside this window yields post-treatment-free devices with $\approx 26\%$ efficiency, improved operational stability, and tight device statistics across multiple chemistries and processing modes. Because TI can be measured during coating and annealing, it converts molecular design rules into actionable process control, providing a portable metric for additive screening and quality assurance in perovskite processing. Furthermore, we validate the practical scalability of this concept by integrating in-situ TI monitoring into a slot-die coating platform. Our results confirm that the coherent transformation mechanism remains fully applicable during large-area deposition, directly guiding the successful fabrication of mini-modules.

Results

In-situ observation of coherent transformation

We select the FA-based perovskite with an n-i-p device structure of fluorine doped tin oxide (FTO)/ SnO_2 /perovskite/2,2',7,7'-tetrakis-(*N,N*-di-4-methoxyphenylamino)-9,9'-spirobifluorene (Spiro-OMeTAD)/Au as a model system for our study, which has demonstrated high photovoltaic performance (Supplementary Table 1). The perovskite layer was fabricated using a 1.5 M precursor solution with the composition of $\text{FA}_{0.98}\text{MA}_{0.02}\text{Pb}(\text{I}_{0.98}\text{Br}_{0.02})_3$. Consistent with previous reports^{30–32}, the control perovskite film exhibits relatively small crystallites and disordered morphology (Supplementary Note 1), resulting in power conversion efficiencies (PCEs) of around 23% with modest stability (Supplementary Fig. 2, Supplementary Table 1, and Supplementary Note 2). To make comparison, we then used two strategies to enhance perovskite film quality, additive engineering using 2-amino-5-fluoropyridine (FPyA), and single-crystal redissolution (SCR) respectively. We note, since we focus on the relationship of the crystallization process and the resulting quality of perovskite films, we did not apply any post-treatment to the perovskite films. For additive engineering with FPyA, the resulting films displayed larger and more ordered crystallites and reduced trap densities of $7.67 \times 10^{14} \text{ cm}^{-3}$ (Supplementary Fig. 1 and Supplementary Note 1), leading to a PCE increase to 26.40% (certified PCE = 25.43%, Supplementary Fig. 3) and enhanced stability (Supplementary Fig. 2 and Supplementary Note 2). The SCR strategy also demonstrated comparable performance, achieving reduced trap densities of $1.15 \times 10^{15} \text{ cm}^{-3}$, a PCE of 25.82% and enhanced stability (Supplementary Figs. 4–5 and Supplementary Notes 3–4). Notably, these PCEs and stabilities are among the highest for perovskite films without post-treatment with a regular device structure^{33–35}, indicating the quality of our perovskite films can be comparable with the state of the art. Below, we then focus on the mechanism of the formation of these high-performance perovskite films.

To elucidate the underlying mechanisms responsible for the enhanced performance and film quality for the two strategies, we employed home-built in-situ absorption spectroscopy. This technique, featuring a high time resolution of 10 ms and a spectral resolution of 0.79 nm, enabled us to capture the rapid evolution of chemical species throughout the crystallization process, with the aim of uncovering the common principles governing high-efficiency perovskite formation regardless of the specific synthetic route. Across all strategies, annealing-time-dependent absorption spectra revealed dramatic spectral shifts within the initial 20 seconds, with the absorption onset

moving rapidly from ~ 500 to 800 nm (Fig. 1a–c). Intriguingly, and unexpectedly for nominally 3D perovskites, absorption features corresponding to layered perovskite structures (n_1, n_2, n_3, n_4) emerged in all films within the first 10 seconds of annealing. These features, observed at $\sim 518 \text{ nm}$ ($n=1$ like layered perovskites, denoted as n_1), $\sim 567 \text{ nm}$ (n_2), $\sim 606 \text{ nm}$ (n_3), and $\sim 642 \text{ nm}$ (n_4), correspond to excitonic absorption peaks of layered perovskites with varying numbers (n) of inorganic $[\text{PbI}_6]$ octahedral layers^{36,37}. This observation suggests a universal transient layered perovskite intermediate phase during 3D perovskite crystallization, intrinsic to FA-based perovskite formation, regardless of the specific strategy employed. However, the dynamics of the emergence, growth, and transformation of these layered perovskite features diverge significantly, resulting in final perovskite films with various qualities, as we will discuss below.

In the control film, the time-dependent absorption intensity curve for the n_1 component exhibits a noticeable plateau (with its first derivative approaches zero) between 5.6 and 7.2 seconds (Fig. 1d and Supplementary Fig. 6a). During this period, the first derivatives of the time-dependent absorption intensity curves of the n_2, n_3 , and n_4 components show simultaneous peaks, indicating the consumption of n_1 species and their transformation into higher- n perovskite species. Subsequently, the absorption intensities of n_1, n_2, n_3 , and n_4 reach their maximum values almost simultaneously, with their first derivatives concurrently approaching zero (Supplementary Fig. 6a). This observation suggests a near-simultaneous growth of various layered perovskites in the control film, even though high- n perovskites are converted from n_1 species. Moreover, the characteristic absorption spectrum of 3D perovskites already appears before 9.5 s, indicating the concurrent formation of layered and 3D perovskites, which contributes to the disordered nature of the perovskite film (Supplementary Fig. 1 and Supplementary Note 1). In contrast, the additive sample follows a markedly different transformation pathway. As shown in Fig. 1e, the time-dependent absorption intensity curves reveal that each layered perovskite reaches its maximum value sequentially, from low- n to high- n layered perovskites, with their first derivatives reaching zero sequentially as well (Supplementary Fig. 6b). This result suggests a collective and orderly layer-by-layer transformation in the additive sample. Notably, the characteristic absorption spectrum of 3D perovskites appears only after the saturation of the high- n absorption, further indicating an orderly transformation of high- n layered perovskites to 3D perovskites. As shown in Fig. 1a, b, once the layered perovskites begin transforming into 3D perovskites, their excitonic absorption peaks disappear, and the entire spectra exhibit the characteristic absorption of 3D perovskites. Moreover, since the absorption coefficient of the 3D perovskite is significantly lower than that of the layered perovskites³⁸, the overall intensity of the absorption spectra decreases during the transformation (Fig. 1a, b). Therefore, the stabilization of the absorption spectra indicates the completion of the transformation from layered perovskites to 3D perovskites. Interestingly, the absorption spectra of additive sample stabilize at 17 s, compared to 12 s for the control sample. This suggests that the critical role of the additive during annealing is to induce a coherent, layer-by-layer, and slower growth process that promotes the ordered crystallization of 3D perovskites. This finding can be further verified by performing similar measurements with various additives reported in the literature for high performance 3D perovskites based solar cells (Supplementary Figs. 8 and 9)^{16,17,19,31,39,40}. Remarkably, we find that the SCR strategy also leads to a collectively and orderly layer-by-layer transformation during the crystallization process, as shown in Fig. 1c, f. Interestingly, the absorption spectra of SCR sample stabilize at 12 s, which is similar to the control sample (Supplementary Fig. 6c), indicating that a collectively layer-by-layer transformation is not necessarily due to a slow growth process, but other more fundamental mechanisms.

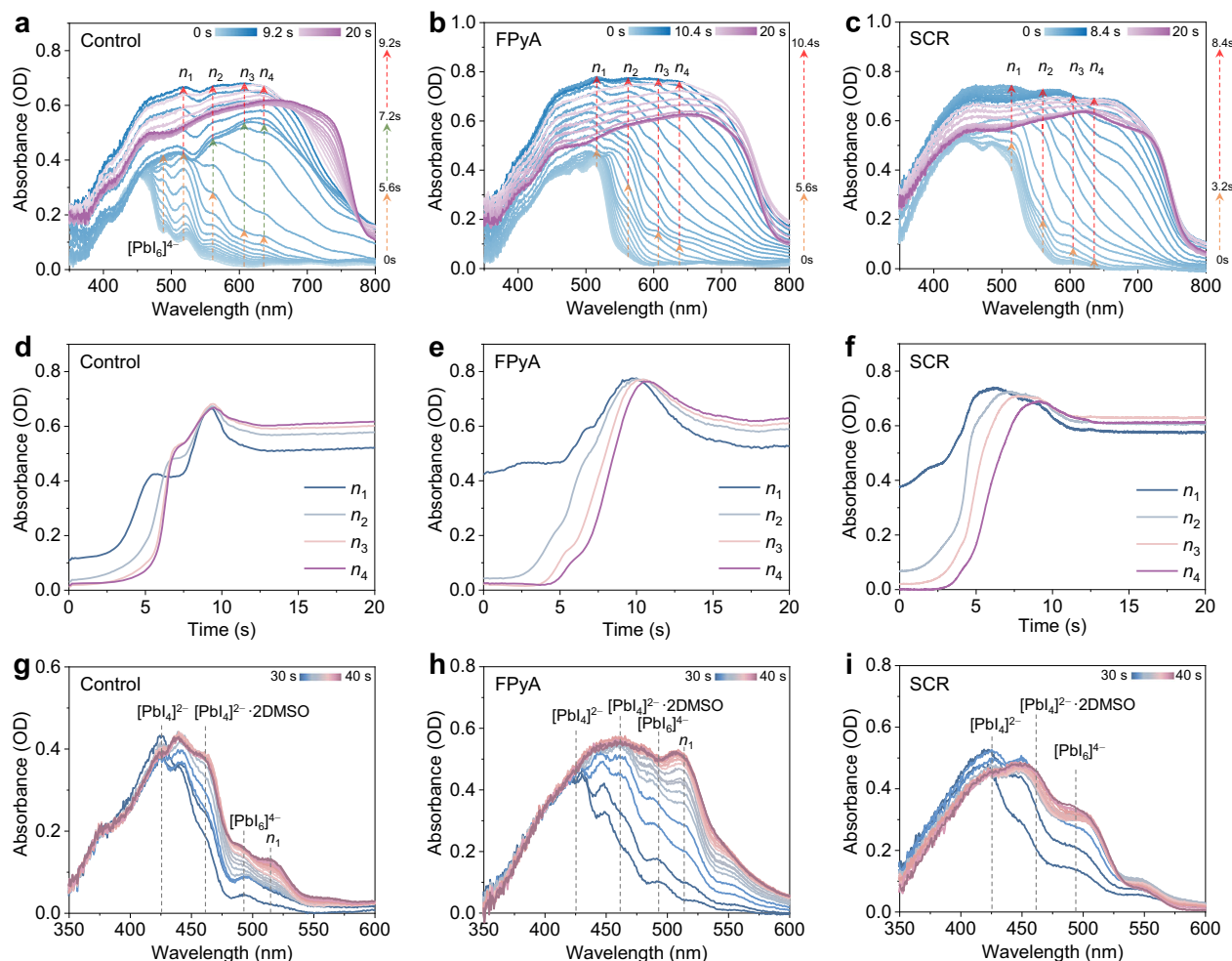


Fig. 1 | In-situ UV-Vis absorption spectroscopy reveals dynamic crystallization under different strategies. **a–c** Annealing-time-dependent absorption spectra for control, FPyA additive, single-crystal redissolution (SCR), showing perovskite phase evolution (extracted from Supplementary Fig. 6a–c). **d–f** Evolution of absorption

intensities for layered perovskite species (n_1 to n_4) during annealing, quantified from spectra in (a–c). **g–i** Spin-coating-time-dependent spectra capturing early-stage film formation (extracted from Supplementary Fig. 7a). n_1 , n_2 , n_3 , n_4 denote layered perovskites. OD, optical density.

Mechanism of single nucleation pathway

To gain further insight into the origins of the different transformation behaviors, we measured the in-situ absorption spectra of the samples during the spin-coating process (Supplementary Fig. 7). The primary differences emerge after the antisolvent dripping step, which occurs at 30 s. For the control perovskite films, the main absorption peaks appear at 425 nm and 460 nm, corresponding to the absorption of $[\text{PbI}_4]^{2-}$, and $[\text{PbI}_4]^{2-} \cdot 2\text{DMSO}$ complexes, respectively (Fig. 1g)^{41,42}. In contrast, the additive significantly enhances the formation of $[\text{PbI}_6]^{4-}$ (490 nm) and n_1 (518 nm) species (Fig. 1h)^{36,37}. X-ray diffraction (XRD) patterns and optical microscopy images reveal that the additive effectively suppresses the formation of the undesirable $\text{MA}_2\text{Pb}_3\text{I}_8 \cdot 2\text{DMSO}$ intermediate phase due to its strong coordination ($\text{N} \rightarrow \text{Pb}^{2+}$) and hydrogen bonding ($\text{F} \cdots \text{MA}^+$) with perovskite precursors^{14,43}. This suppression leads to abundant $[\text{PbI}_6]^{4-}$, and establish a single nucleation pathway and rapid nucleation of the n_1 phase (Supplementary Fig. 10 and Supplementary Note 5). We believe that the establishment of a single nucleation pathway is essential for the coherently layer-by-layer transformation of perovskites during the annealing process.

Similarly, the SCR strategy achieves coherent transformation of layered perovskites during annealing by establishing the same initial state after spin-coating (Fig. 1i). This state is characterized by the rapid and dominant formation of pre-nucleated $[\text{PbI}_6]^{4-}$ (490 nm) and n_1

(518 nm) immediately after antisolvent dripping. Inherited from the perovskite single-crystal structure in the precursor, this pre-nucleation bypasses undesirable intermediate phases prevalent in the control sample. The resulting iodide-rich environment and abundant n_1 nuclei act as seeds, guiding an ordered, layer-by-layer transition of layered perovskites into the desired 3D structure during subsequent annealing. In this case, even the nucleation and growth are much faster than the control sample, very high-quality perovskite films can still be achieved through the coherent transformation. Therefore, this observation confirms the importance of establishing a single nucleation pathway for the coherently layer-by-layer transformation of perovskites.

Interestingly, we can further confirm that the role of a single nucleation pathway for the coherently layer-by-layer transformation in the control sample is achieved by low-temperature spin-coating (LTC). DFT calculations show that the reaction energy of the detrimental $\text{MA}_2\text{Pb}_3\text{I}_8 \cdot 2\text{DMSO}$ phase is higher than $[\text{PbI}_6]^{4-}$ (Supplementary Note 6). At a reduced temperature of -10°C , the formation kinetics of the $\text{MA}_2\text{Pb}_3\text{I}_8 \cdot 2\text{DMSO}$ phase are drastically suppressed while $[\text{PbI}_6]^{4-}$ can still grow, as shown in XRD (Supplementary Fig. 11a, b), optical microscopy (Supplementary Fig. 11c, d) and in-situ absorption spectrum measurements (Supplementary Figs. 12–14, and Supplementary Note 7). Remarkably, this film also shows a coherently layer-by-layer transformation during annealing process (Supplementary

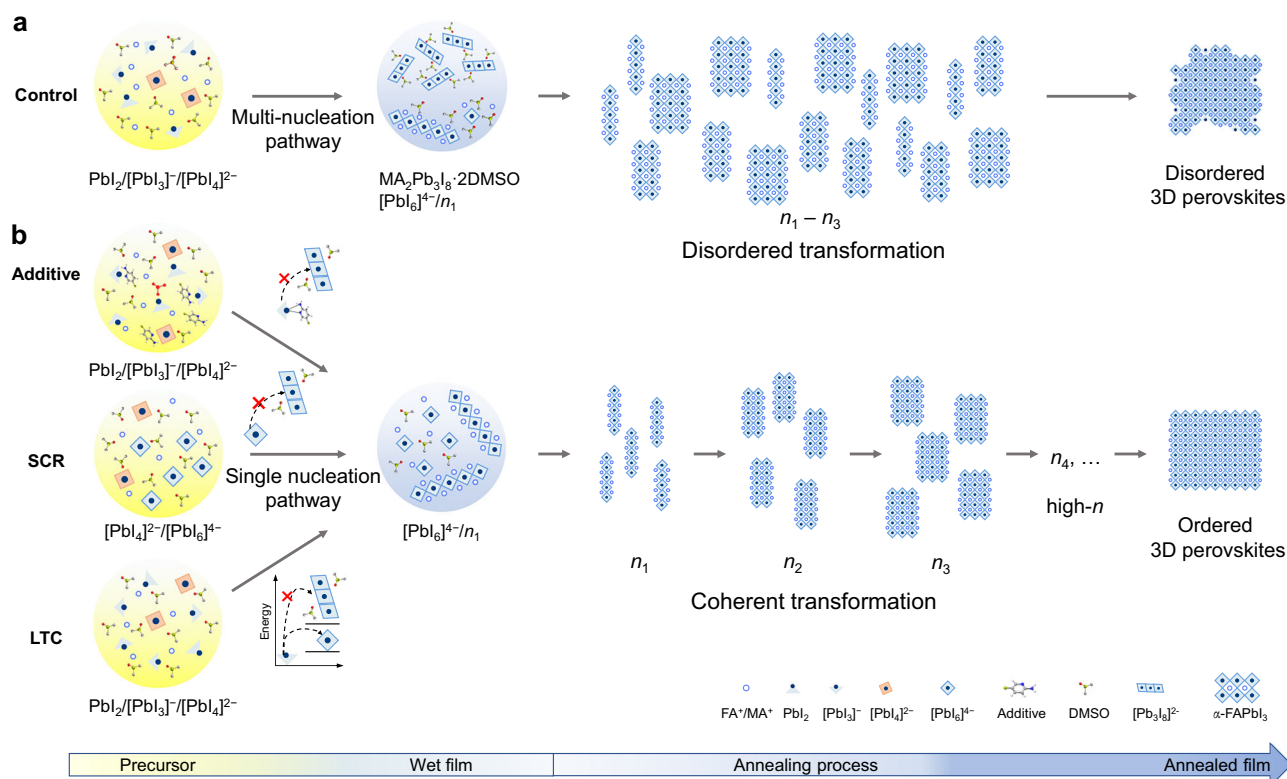


Fig. 2 | Schematic diagram illustrating distinct crystallization pathways and the principle of coherent transformation in perovskite films. a Crystallization pathway in the control film. The process is characterized by rapid, uncontrolled solvent evaporation and concurrent formation of various disordered intermediate phases, including the detrimental $\text{MA}_2\text{Pb}_3\text{I}_8 \cdot 2\text{DMSO}$. This leads to rapid, non-sequential nucleation and growth of perovskite species and a disordered transformation from intermediate/low- n phases to the 3D structure, resulting in a polycrystalline film with high defect density. **b** Crystallization pathways facilitated by strategies promoting coherent transformation (Additive Engineering, Single-Crystal Redissolution, and Low-Temperature Crystallization). These strategies guide the crystallization towards a coherent layer-by-layer transformation. These strategies establish a favorable environment for the coherent emergence and

transition of layered perovskites into the desired 3D structure by optimizing mechanisms such as suppressing detrimental intermediates and promoting beneficial nuclei ($[\text{PbI}_6]^{4-}$, n_1) through additive engineering, providing pre-formed iodide-rich species and n_1 seeds through single-crystal redissolution, and retarding kinetics via low-temperature crystallization to favor controlled conversion. This ordered process results in a highly crystalline film with significantly reduced defect density, underpinning the enhanced photovoltaic performance. Colors indicate processing stages: yellow, precursor solution; light blue, wet film after spin-coating; blue gradient, annealing process. Red crosses denote suppressed intermediate phases. LTC, low-temperature crystallization. The energy diagram illustrates the reaction energy barrier for the formation of the intermediate phase.

Note 7), and resulting in low trap-density films ($9.82 \times 10^{14} \text{ cm}^{-3}$) and high PCE devices (25.74%) (Supplementary Figs. 15–16 and Supplementary Note 8). Therefore, we now can safely conclude that in these seemingly very different approaches to achieve high quality perovskite films, the common key is to induce a single nucleation pathway for the coherently layer-by-layer transformation, as schematically shown in Fig. 2.

Quantification by transformation index

We then try to establish a correlation between the extent of coherent transformation and the resulting film quality. Based on the insights from in-situ absorption spectroscopy regarding the dynamics of layered perovskite evolution (Fig. 1d–f), we introduce a metric termed the “transformation index (TI)”. This index is calculated by multiplying the time interval between the absorption intensity maxima of n_1 and n_4 with the time interval for the transition from layered to 3D perovskites (Supplementary Note 9). We first explored the impact of additive engineering on coherent transformation and device performance by selecting various additives with different predicted binding energies (E_b) to uncoordinated Pb^{2+} ions (Fig. 3a, Supplementary Fig. 17, and Supplementary Note 10). In-situ absorption spectra reveal a general trend where stronger interactions between additives and PbI_2 initially promote a more ordered transformation process as reflected in the TI (Supplementary Fig. 18–19). Figure 3a shows that the TI is positively

correlated with binding energies. For instance, the TI for the control film is 0.57 s^2 , while for the FPyA-based film (representing an optimized interaction strength), it is 7.23 s^2 . Time-Resolved Photoluminescence (TRPL) measurements reveal that increasing TI leads to a significant reduction in defect state density, from approximately 10^{15} cm^{-3} to 10^{14} cm^{-3} (Fig. 3a and Supplementary Fig. 20). This positive correlation between TI and low defect density holds up to an optimal interaction strength. In the case of 2-aminopyridine, the excessively strong interaction with Pb^{2+} hinders the formation of perovskite nuclei (n_1) (Supplementary Fig. 18). Consequently, the transformation process is hardly observable and its TI is very large, leading to an abundance of the yellow phase perovskites (Supplementary Fig. 21), high defect state density, and consequently poor device performance (Fig. 3a). The existence of an optimal additive interaction strength and TI is consistent with the scenario of single nucleation pathway induced coherently layer-by-layer transformation, since the weak interaction is not strong enough to suppress the formation of $\text{MA}_2\text{Pb}_3\text{I}_8 \cdot 2\text{DMSO}$ phase, and the too strong interaction will hinder the formation of nuclei. Additionally, the optimal interaction strength for the coherent transformation aligns with the improved device efficiencies and operation stability, as shown in Fig. 3a and Supplementary Figs. 22 and 23. Notably, along with the high efficiency, in the 85°C thermal stability measurement and maximum power point tracking (MPPT) measurements, FPyA-based devices consistently maintained

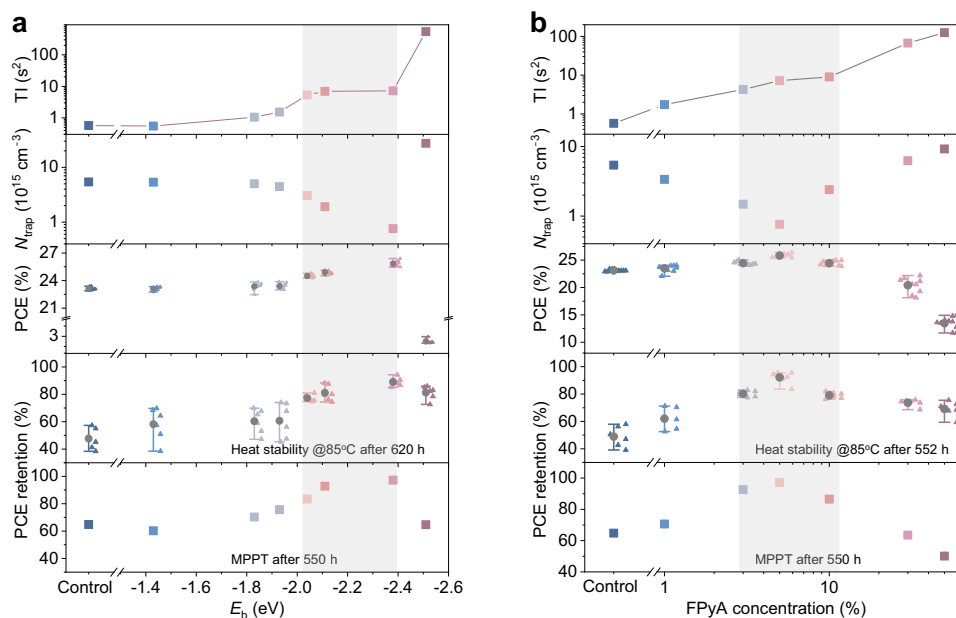


Fig. 3 | Impact of coherent transformation extent on perovskite crystallization and devices performance. **a** Plots of binding-energy (E_b , different additives with uncoordinated Pb^{2+} ions)-dependent transformation index (TI), device efficiency (PCE), and stability. **b** Plots of additive concentration-dependent TI, defect density (N_{trap}), device efficiency, and stability. FPyA concentration is varied from 1 to 50 mol% relative to the lead concentration. For efficiency statistics, 10 devices are used for each group. For 85 °C thermal stability, 6 unencapsulated devices are used for each group, with efficiency retention values obtained after approximately 600 h of aging. For efficiency statistics and 85 °C thermal stability, data are presented as

box plots overlaid with individual data points (triangles); gray circles denote the mean values. The upper and lower whiskers represent the maximum and minimum values, respectively. For MPPT, efficiency retention values of unencapsulated PSCs are obtained after 550 h of aging. The gray shaded regions in **a** and **b** indicate the optimal windows for additive-precursor interaction strength and additive concentration, respectively, where coherent transformation yields the highest device efficiencies and stabilities. Color scale in **a** indicates increasing binding energy from blue (weak) to red (strong). Color scale in **(b)** indicates increasing FPyA concentration from blue (low) to red (high).

best performance, retaining over 90% of their initial average efficiency after approximately 600 hours of aging, which is also among the best stable PSCs without post-treatments^{33,44–46}.

Similar to interaction strength, we find that the concentration of the additive also critically influences TI and film quality, exhibiting a concentration-dependent optimum. (Fig. 3b and Supplementary Figs. 24 and 25). From 1% to 5% FPyA concentrations, the in-situ absorption spectra display a growing trend in TI. However, at concentrations exceeding 10%, the additive begins to hinder the transformation. At a concentration of 50%, the n_1 species no longer transforms to higher- n species. This trend of ordered transformation aligns with decreased defect density, improved device performance and stability (Fig. 3b and Supplementary Figs. 26–28). This concentration-dependent behavior again illustrates the mechanism of single nucleation pathway induced coherently layer-by-layer transformation, since the low concentration additive is not enough to suppress the formation of $\text{MA}_2\text{Pb}_3\text{I}_8 \cdot 2\text{DMSO}$ phase, and the excess high concentration leads to detrimental over-coordination with Pb^{2+} ions that stagnates the structural rearrangement. At optimal concentrations, however, additives (even bulky ones) are dynamically expelled to grain boundaries without impeding the phase transition^{16,40}. This explicitly reinforces the existence of a Sabatier optimum for both interaction strength and concentration. Unlike interaction strength, the importance of additive concentration is rarely discussed for optimizing perovskite films in literature^{47,48}, our findings emphasize that both are critical for controlling coherent transformation and maximizing device performance.

Validation of optimal transformation window

From the above studies (Fig. 3), we find that an optimum TI is within the range of 4–12 s^2 for high quality perovskite films. This result is consistent with SCR and low temperature spin-coating process, where

TIs are 9.52 s^2 and 7.98 s^2 , respectively. Crucially, this optimal TI window is largely insensitive to typical variations in heating conditions, maintaining its validity across different substrate thicknesses and minor annealing temperature fluctuations (Supplementary Figs. 29–31 and Supplementary Table 2). To further validate our findings, we investigated a diverse range of additives, including both additives developed in our lab (all showing PCEs > 25% without post-treatment, Supplementary Figs. 32–34, and Supplementary Table 3) as well as previously reported additives for high-performance PSCs in the literature (Supplementary Figs. 8 and 9 and Supplementary Table 4)^{16,17,19,31,39,40}. The results consistently demonstrate that these additives induce a similar coherent and ordered transformation from layered perovskites to 3D perovskites during crystallization (Supplementary Table 5), and TIs for high-performance additives fall into the range of 4–12 s^2 . Interestingly, we find that the relationship between the transformation index and device PCEs indeed exhibits a volcano shape, indicative of a Sabatier principle-guided interaction (Fig. 4). Consistently, Fig. 4 also suggests that the defect density shows an inverse correlation, reaching its minimum within the optimal TI range associated with peak PCEs. Moving beyond lab-scale spin-coating, we successfully integrated in-situ TI monitoring into a slot-die coating platform; the coherent transformation was perfectly preserved (TI = 4.83 s^2), which guided the fabrication of 5 cm × 5 cm mini-modules (Supplementary Figs. 35 and 36, Supplementary Table 2, and Supplementary Movie 1), demonstrating its broad applicability for scalable processing.

Discussion

In summary, we demonstrate that the coherent transformation of layered intermediates into three-dimensional perovskites during crystallization constitutes a unifying mechanism underlying diverse optimization strategies. This transformation is driven by a single,

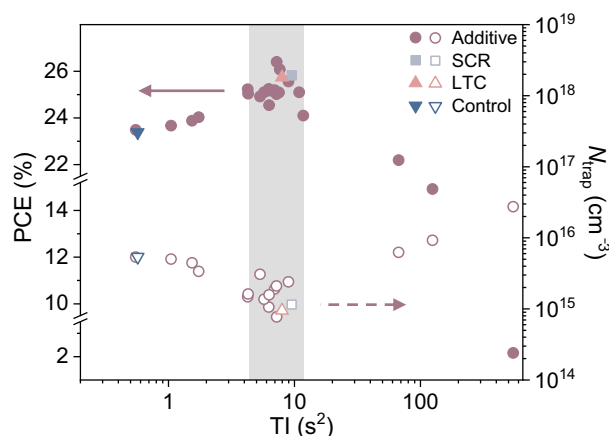


Fig. 4 | Plot of PCE and defect density versus transformation index. PCE, defect density, and transformation index values were obtained from this work and literature (Supplementary Table S)^{16,17,19,31,39,40}. Filled and hollow symbols correspond to PCE and trap density (N_{trap}), respectively. The symbols represent different strategies: Circles denote the additive engineering strategy (including FPyA at various concentrations, additives with varying binding energies from this work, and previously reported additives for high-performance PSCs); squares represent the single-crystal redissolution strategy; upward triangles signify the low-temperature crystallization strategy; and downward triangles represent the control samples. The optimal TI range falls within 4–12 s^2 , yielding highly efficient devices.

ordered nucleation pathway, and its extent, quantified by the transformation index, directly governs the film microstructure, trap density, and, consequently, device efficiency and stability. Framing crystallization in terms of nucleation pathway control and the Sabatier optimum yields a quantitative, generalizable principle for rational perovskite processing. The transformation index and its associated inline metrics provide practical levers for process control and monitoring, as demonstrated in scalable slot-die coating, with broader implications for the controlled synthesis of high-quality polycrystalline optoelectronic materials beyond perovskite photovoltaics.

Methods

Materials

Lead iodide (PbI_2 , 99.99%) and lead bromide (PbBr_2 , $\geq 98\%$), bis(trifluoromethane)sulfonimide lithium salt (Li-TFSI), 4-tertbutyl-pyridine (TBP) and 4-isopropyl-4'-methyldiphenyliodonium tetrakis(pentafluorophenyl)borate (TPFB) were purchased from TCI Shanghai (China). Methylammonium chloride (MACl, $\geq 99.5\%$), Methylammonium bromide (MABr, $\geq 99.5\%$), 2,2',7,7'-tetrakis [N,N -di(4-methoxyphenyl)amine]-9,9'-spirobifluorene (Spiro-OMeTAD, $>99.8\%$), FK209-Co(III)-TFSI ($\geq 99\%$) and poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA) were purchased from Xi'an Ploymer Light Technology Corp. Formamidinium iodide (FAI, $\geq 99.9\%$) was purchased from Youxuan Tech, China. Chlorobenzene (CB, anhydrous), N,N -dimethylformamide (DMF, anhydrous), dimethyl sulfoxide (DMSO, anhydrous), acetonitrile (ACN), Ethyl acetate (EA, anhydrous), stannous chloride dihydrate ($\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$, 99.99%), thioglycolic acid (TGA, 98%), and urea ($\geq 99.5\%$) were purchased from Sigma-Aldrich. All the materials were used without further purification.

Perovskite precursor solution

(FAPbI_3)_{0.98}(MAPbBr_3)_{0.02}: The precursor solution of perovskite was prepared by dissolving PbI_2 , FAI, MACl, MABr, PbBr_2 , and FPyA with a molar ratio of 1.09:1:0.35:0.02:0.02:0.051 at 52 wt% in a mixed solution of DMF and DMSO ($v/v = 8:1$). Additives with varying binding energies to uncoordinated Pb^{2+} were dissolved with a corresponding molar concentration relative to lead concentration in the perovskite precursors. The obtained precursor solution was then stirred at room

temperature overnight and filtered through 0.22 μm polytetrafluoroethylene filters before use. The SCR perovskite precursor of (FAPbI_3)_{0.98}(MAPbBr_3)_{0.02} was prepared by dissolving 868.45 mg FAPbI_3 crystals and 13.24 mg MAPbBr_3 crystals and 33.08 mg MACl in DMF/DMSO (8:1, V/V) solvents.

Solution and film characterization

XPS measurements were conducted by using a Thermo Scientific K-Alpha+ XPS spectrometer with an Al $K\alpha$ anode ($E = 1486.6 \text{ eV}$). The pressure in the analysis chamber was maintained at $\approx 5 \times 10^{-9}$ mbar. To prevent exposure to the atmosphere, all perovskite films were transferred into the XPS chamber through a glovebox. NMR measurements were conducted using a Bruker 400 MHz. XRD data were collected using a Bruker D8 Advance diffractometer. Optical micrographs were obtained with an LV100ND Nikon microscope. Steady-state absorbance spectra were measured by a UV-vis spectrophotometer with an integrating sphere (PerkinElmer, Lambda 950). SEM images were obtained using a JEOL JSM-7610F Plus instrument. For PLQE measurements, a three-step technique was used through the combination of a 445-nm CW laser, an optical fiber, a spectrometer, and an integrating sphere.

Film and device fabrication

FTO glasses were washed with dish soap, water, and ethanol, followed by ultrasonic cleaning for 15 min. Then, they were dried using a nitrogen gun and further treated with O_2 plasma for 15 min. After depositing a compact stannic oxide (SnO_2) film on FTO glass via chemical bath deposition (CBD): 10 mM $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ in ethanol was mixed with thioglycolic acid (TGA) as a complexing agent, heated at 70 $^\circ\text{C}$ for 2 hours, followed by rinsing with deionized water and annealing at 180 $^\circ\text{C}$ for 30 min⁴⁹. A diluted potassium chloride (KCl) solution was spin-coated on FTO- SnO_2 substrates at 3000 rpm for 30 s, followed by annealing at 110 $^\circ\text{C}$ for 10 min in air. The substrates were then transferred into an air glovebox. Perovskite precursor solution was spin-coated onto the substrate at 1000 rpm for 10 s and 5000 rpm for 30 s. At 10 seconds before the end of spin-coating, 90 μL of ethyl acetate was deposited onto the substrate. A spin coater with adjustable chamber temperature was used for perovskite film preparation. For the -10°C processing condition, the temperature parameter was set to -10°C cooling mode. Both the substrate and precursor solution were pre-equilibrated in the spin coater chamber for 10 minutes prior to deposition. The resulting film was then placed on a hotplate at 100 $^\circ\text{C}$ for 60 min. The Spiro-OMeTAD solution was prepared by dissolving 30 μL of TBP, 18 μL of Li-TFSI solution (520 mg in 1 mL acetonitrile), 29 μL of FK209-Co(III)-TFSI solution (300 mg in 1 mL acetonitrile) and 73 mg of Spiro-OMeTAD in 1 mL CB. The PTAA solution was prepared by dissolving 30 mg of PTAA and 3 mg of TPFB in 1 mL CB. The solution was then spin-coated on the perovskite films at 3000 rpm (Spiro-OMeTAD) or 2000 rpm (PTAA) for 30 s, respectively. Finally, an 80-nm thick of gold layer was deposited using thermal evaporation.

Device characterization

All perovskite solar cells were measured under simulated AM1.5 G irradiation at 100 mW cm^{-2} using a solar simulator (Class AAA, SS-F5-3A, Enlitech), with current–voltage curves were recorded by a Keithley 2450 source meter. The light intensity was calibrated using a Newport standard silicon cell 91150. Current density–voltage (J - V) characteristics were measured in N_2 using a scan rate of 0.02 V s^{-1} . No preconditioning was applied to the devices before measurements. IPCE measurements were conducted using a QE-R-900AD system (Nanjing Ouyi Optoelectronics Technology). An LED was utilized as a bias light source to simulate the device operating conditions (~ 0.2 Sun). A Hamamatsu S1337-1010BQ silicon diode used for IPCE measurements was calibrated at the National Institute of Metrology, China. Steady-state power output measurements were carried out at a fixed voltage near

the maximum power point identified in the J - V curve (Supplementary Fig. 37). For the moisture tolerance test, unencapsulated perovskite solar cells using Spiro-OMeTAD were stored in ambient air of 60% ($\pm 5\%$) and 20% ($\pm 5\%$) relative humidity in the dark. For thermal stability measurements, unencapsulated perovskite solar cells using PTAA were kept in a N_2 glove-box under heat stress of 85°C. In the maximum power point tracking measurement, unencapsulated devices with PTAA were kept in N_2 at approximately 45°C and monitored using an MPP Tracking-4B system (Shenzhen Lancheng Technology Ltd.) under LED-based Sunlight Simulators (Newport Verasol) with an emission spectrum of 400–1000 nm.

In-suit UV-vis absorption spectroscopy measurement

In-suit absorption spectra were collected by an ISAS-HI001 system with a high temporal resolution of 10 ms and a spectral resolution of 0.79 nm (Nanjing Ouyi Optoelectronics Technology). Measurements were performed using an FTO/SnO₂ substrate. The real-time absorbance (A) of the perovskite film during the spin-coating and thermal annealing processes was calculated using the equation $A = \lg(I_0/I)$, where I_0 and I represent the intensities of the reflected light from the bare substrate and the perovskite-coated substrate, respectively.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data generated in this study are provided in the article and Supplementary Information, and the raw data supporting this study are available from the Source Data file. Source data are provided with this paper.

References

- Park, N.-G., Grätzel, M., Miyasaka, T., Zhu, K. & Emery, K. Towards stable and commercially available perovskite solar cells. *Nat. Energy* **1**, 16152 (2016).
- Han, T.-H. et al. A roadmap for the commercialization of perovskite light emitters. *Nat. Rev. Mater.* **7**, 757–777 (2022).
- Li, M. et al. Acceleration of radiative recombination for efficient perovskite LEDs. *Nature* **630**, 631–635 (2024).
- Min, H. et al. Additive treatment yields high-performance lead-free perovskite light-emitting diodes. *Nat. Photonics* **17**, 755–760 (2023).
- Wei, H. & Huang, J. Halide lead perovskites for ionizing radiation detection. *Nat. Commun.* **10**, 1066 (2019).
- Yang, G. et al. Towards efficient, scalable and stable perovskite/silicon tandem solar cells. *Nat. Photonics* **19**, 913–924 (2025).
- Jung, M., Ji, S.-G., Kim, G. & Seok, S. I. Perovskite precursor solution chemistry: from fundamentals to photovoltaic applications. *Chem. Soc. Rev.* **48**, 2011–2038 (2019).
- Zhu, P. et al. Aqueous synthesis of perovskite precursors for highly efficient perovskite solar cells. *Science* **383**, 524–531 (2024).
- Sánchez, S., Pfeifer, L., Vlachopoulos, N. & Hagfeldt, A. Rapid hybrid perovskite film crystallization from solution. *Chem. Soc. Rev.* **50**, 7108–7131 (2021).
- Maschwitz, T. et al. How crystallization additives govern halide perovskite grain growth. *Nat. Commun.* **16**, 9894 (2025).
- Jeong, J. et al. Pseudo-halide anion engineering for α -FAPbI₃ perovskite solar cells. *Nature* **592**, 381–385 (2021).
- Zhu, L. et al. Unveiling the additive-assisted oriented growth of perovskite crystallite for high performance light-emitting diodes. *Nat. Commun.* **12**, 5081 (2021).
- Li, C. et al. Rational design of Lewis base molecules for stable and efficient inverted perovskite solar cells. *Science* **379**, 690–694 (2023).
- Huang, Z. et al. Anion- π interactions suppress phase impurities in FAPbI₃ solar cells. *Nature* **623**, 531–537 (2023).
- Xu, J. et al. Anion optimization for bifunctional surface passivation in perovskite solar cells. *Nat. Mater.* **22**, 1507–1514 (2023).
- Li, M. et al. Orientated crystallization of FA-based perovskite via hydrogen-bonded polymer network for efficient and stable solar cells. *Nat. Commun.* **14**, 573 (2023).
- Wang, F. et al. Monolithically-grained perovskite solar cell with mortise-tenon structure for charge extraction balance. *Nat. Commun.* **14**, 3216 (2023).
- Fu, S. et al. On-demand formation of Lewis bases for efficient and stable perovskite solar cells. *Nat. Nanotechnol.* **20**, 772–778 (2025).
- Park, J. et al. Controlled growth of perovskite layers with volatile alkylammonium chlorides. *Nature* **616**, 724–730 (2023).
- Lin, X. et al. Degradable additive couple enable pure and stable α -phase FAPbI₃ for perovskite solar cells. *Adv. Mater.* **37**, 2418008 (2025).
- Miao, Y. et al. Green solvent enabled scalable processing of perovskite solar cells with high efficiency. *Nat. Sustainability* **6**, 1465–1473 (2023).
- Kim, S.-G. et al. How antisolvent miscibility affects perovskite film wrinkling and photovoltaic properties. *Nat. Commun.* **12**, 1554 (2021).
- Wang, Y. et al. Solvent-assisted reaction for spontaneous defect passivation in perovskite solar cells. *Nat. Photonics* **19**, 985–991 (2025).
- Kim, Y. Y. et al. Roll-to-roll gravure-printed flexible perovskite solar cells using eco-friendly antisolvent bathing with wide processing window. *Nat. Commun.* **11**, 5146 (2020).
- Wang, Y. et al. Intermediate phase free α -FAPbI₃ perovskite via green solvent assisted perovskite single crystal redissolution strategy. *Adv. Mater.* **35**, 2302298 (2023).
- Feng, W. et al. Perovskite crystals redissolution strategy for affordable, reproducible, efficient and stable perovskite photovoltaics. *Mater. Today* **50**, 199–223 (2021).
- Zhou, H. et al. Efficient and stable perovskite mini-module via high-quality homogeneous perovskite crystallization and improved interconnect. *Nat. Commun.* **15**, 6679 (2024).
- Chen, M. et al. Low-temperature sequential deposition for efficient inverted perovskite solar cells. *Nat. Commun.* **16**, 5746 (2025).
- Chen, M. et al. “Freezing” intermediate phases for efficient and stable FAPbI₃ perovskite solar cells. *Energy Environ. Sci.* **17**, 3375–3383 (2024).
- Zou, Y. et al. A crystal capping layer for formation of black-phase FAPbI₃ perovskite in humid air. *Science* **385**, 161–167 (2024).
- Wu, Y., Wang, Q., Chen, Y., Qiu, W. & Peng, Q. Stable perovskite solar cells with 25.17% efficiency enabled by improving crystallization and passivating defects synergistically. *Energy Environ. Sci.* **15**, 4700–4709 (2022).
- Ma, Y. et al. Bi-molecular kinetic competition for surface passivation in high-performance perovskite solar cells. *Energy Environ. Sci.* **17**, 1570–1579 (2024).
- Kong, J. et al. CO₂ doping of organic interlayers for perovskite solar cells. *Nature* **594**, 51–56 (2021).
- Li, S. et al. High-efficiency and thermally stable FACsPbI₃ perovskite photovoltaics. *Nature* **635**, 82–88 (2024).
- Gong, C. et al. Silver coordination-induced n-doping of PCBM for stable and efficient inverted perovskite solar cells. *Nat. Commun.* **15**, 4922 (2024).
- Wang, N. et al. Perovskite light-emitting diodes based on solution-processed self-organized multiple quantum wells. *Nat. Photonics* **10**, 699–704 (2016).
- Liang, C. et al. Two-dimensional Ruddlesden–Popper layered perovskite solar cells based on phase-pure thin films. *Nat. Energy* **6**, 38–45 (2020).

38. Papavassiliou, G. C. Three- and low-dimensional inorganic semiconductors. *Prog. Solid State Chem.* **25**, 125–270 (1997).
39. Chen, R. et al. Reduction of bulk and surface defects in inverted methylammonium- and bromide-free formamidinium perovskite solar cells. *Nat. Energy* **8**, 839–849 (2023).
40. Zhang, J. et al. Inhibiting ion migration through chemical polymerization and chemical chelation toward stable perovskite solar cells. *Angew. Chem. Int. Ed.* **62**, e202314106 (2023).
41. Rahimnejad, S., Kovalenko, A., Forés, S. M., Aranda, C. & Guerrero, A. Coordination chemistry dictates the structural defects in lead halide perovskites. *ChemPhysChem* **17**, 2795–2798 (2016).
42. Sanches, A. W. P., Da Silva, M. A. T., Cordeiro, N. J. A., Urbano, A. & Lourenço, S. A. Effect of intermediate phases on the optical properties of PbI_2 -rich $\text{CH}_3\text{NH}_3\text{PbI}_3$ organic–inorganic hybrid perovskite. *Phys. Chem. Chem. Phys.* **21**, 5253–5261 (2019).
43. Feng, W. et al. Small amines bring big benefits to perovskite-based solar cells and light-emitting diodes. *Chem* **8**, 351–383 (2022).
44. Liu, C. et al. Bimolecularly passivated interface enables efficient and stable inverted perovskite solar cells. *Science* **382**, 810–815 (2023).
45. Qu, Z. et al. Enhanced charge carrier transport and defects mitigation of passivation layer for efficient perovskite solar cells. *Nat. Commun.* **15**, 8620 (2024).
46. Jiang, Q. et al. Surface reaction for efficient and stable inverted perovskite solar cells. *Nature* **611**, 278–283 (2022).
47. McMeekin, D. P. et al. Intermediate-phase engineering via dimethylammonium cation additive for stable perovskite solar cells. *Nat. Mater.* **22**, 73–83 (2023).
48. Kim, M. et al. Methylammonium Chloride Induces Intermediate Phase Stabilization for Efficient Perovskite Solar Cells. *Joule* **3**, 2179–2192 (2019).
49. Yoo, J. J. et al. Efficient perovskite solar cells via improved carrier management. *Nature* **590**, 587–593 (2021).

Acknowledgements

P.Y., H.S., and Z.W. contributed equally to this work.

Author contributions

J.W. conceived the idea and designed the experiments. R.L., W.H., and J.W. supervised the work. P.Y., H.S., and X.W. fabricated the perovskite films and devices, and carried out the device characterization. P.Y., H.S., X.W., and S.Y. carried out in-situ UV-vis absorption. P.Y. and S.Y. carried out the TRPL measurement and fitting. P.Y. and H.S. carried out the XRD and optical microscopy measurements. P.Y., S.Y., X.W., and H.S. measured the device stability with the assistance of Bo Xu (B.X.). P.Y., X.W., H.S., B.X., W.L., and X.G. carried out the PLQE, NMR, XPS, and FTIR measurements. Ju.W. and F.W. carried out the DFT calculation. K.L. fabricated the mini-module device. J.W., R.L., P.Y., Z.W., and L.Z. analyzed the data. R.L. wrote the first draft of the manuscript. J.W. and W.H.

provided major revisions. All authors discussed the results and commented on the manuscript.

Funding

This work is financially supported by the National Key Research and Development Program of China (2022YFA1204800), the National Natural Science Foundation of China (62288102), Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China (JYB2025XDXM406).

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-026-75161-y>.

Correspondence and requests for materials should be addressed to Renzhi Li, Wei Huang or Jianpu Wang.

Peer review information *Nature Communications* thanks Lixin Xiao and the other anonymous reviewers for their contribution to the peer review of this work. A peer review file is available.

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026